Lê Anh Thư - 24160043 - Lab04

Bài 1: a) A(10;5) ; B(1;4) ; C(5;8) ; D(9;2) ; E(12;10) ; F(15;8) ; G(7;7)

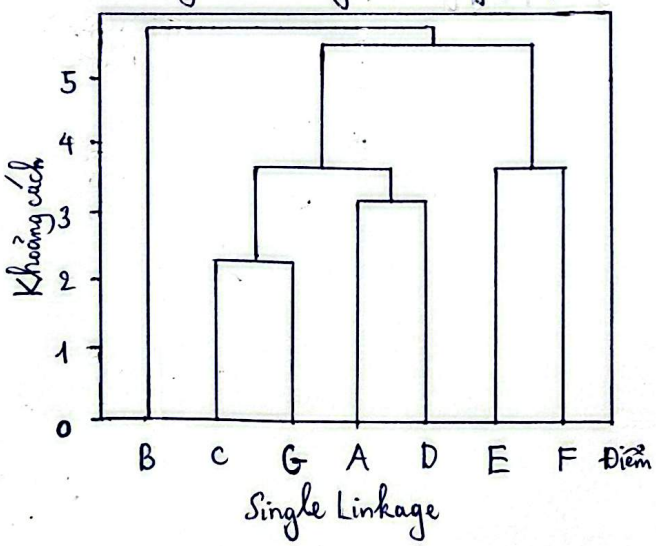
- Tính khoảng cách giữa các điểm, sử dụng công thức:

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

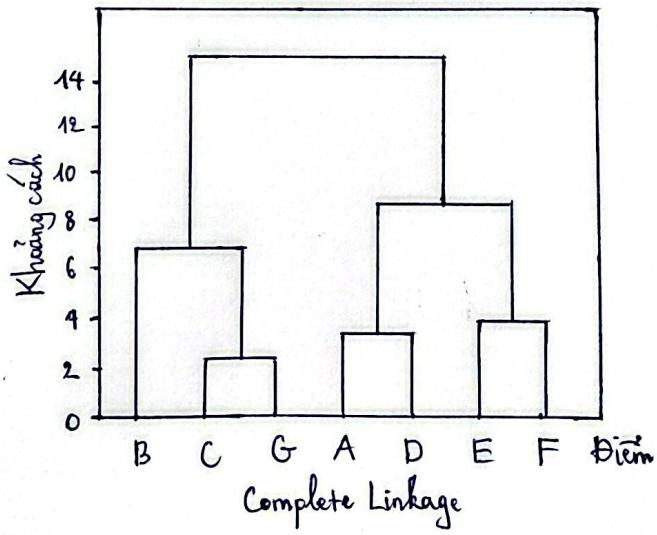
- Ma trận khoảng cách:

	A	B	C	D	E	F	G
A	0	9,06	5,83	3,16	5,39	5,83	3,61
B	9,06	0	5,66	8,25	12,53	14,56	6,71
C	5,83	5,66	0	7,21	7,28	10,00	2,24
D	3,16	8,25	7,21	0	8,54	8,49	5,39
E	5,39	12,53	7,28	8,54	0	3,61	5,83
F	5,83	14,56	10,00	8,49	3,61	0	8,06
G	3,61	6,71	2,24	5,39	5,83	8,06	0

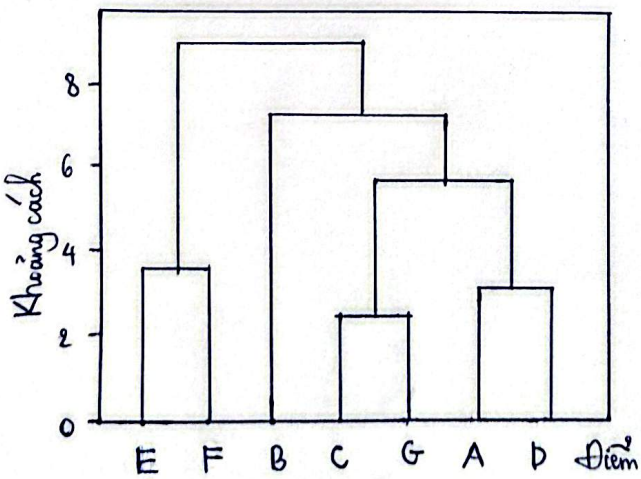
- Vẽ dendrogram bằng 3 phương pháp:



Single Linkage



Complete Linkage



Average Linkage

Được quét bằng CamScanner

b) A(2;3) ; B(5;1) ; C(6;4) ; D(1;7) ; E(4;6)

- Tính khoảng cách giữa các điểm, sử dụng công thức:

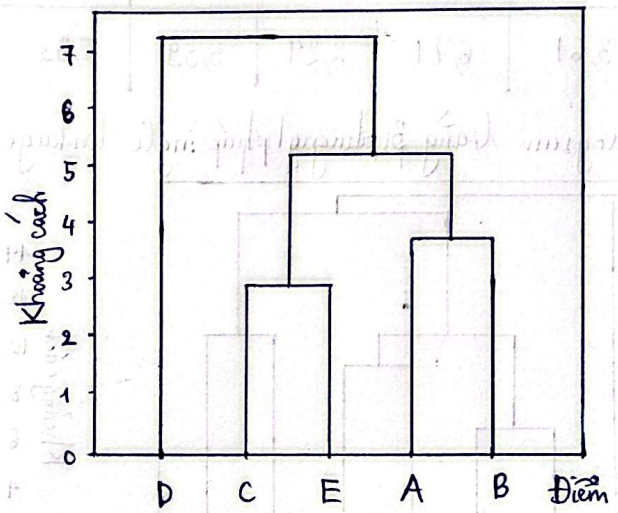
$$d(P_1,P_2) = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

- Ma trận khoảng cách:

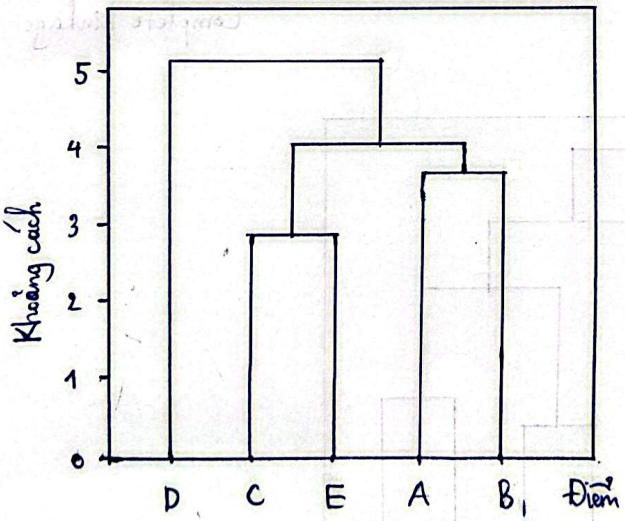
	A	B	C	D	E
A	0	3,61	4,12	4,12	3,61
B	3,61	0	3,61	7,21	5,10
C	4,12	3,16	0	5,83	2,83
D	4,12	7,21	5,83	0	3,16
E	3,61	5,10	2,83	3,16	0

- Vẽ dendrogram bằng 3 phương pháp:

Single Linkage



Complete Linkage



Average Linkage

Được quét bằng CamScanner

c) $P_1(1;2)$; $P_2(3;5)$; $P_3(8;4)$; $P_4(7;1)$; $P_5(2;8)$; $P_6(6;7)$

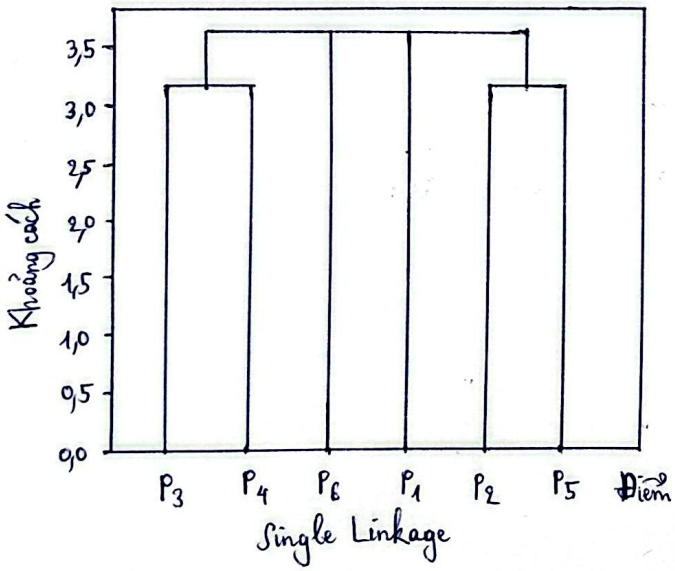
- Tính khoảng cách giữa các điểm, sử dụng công thức:

$$d(P_1, P_2) = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

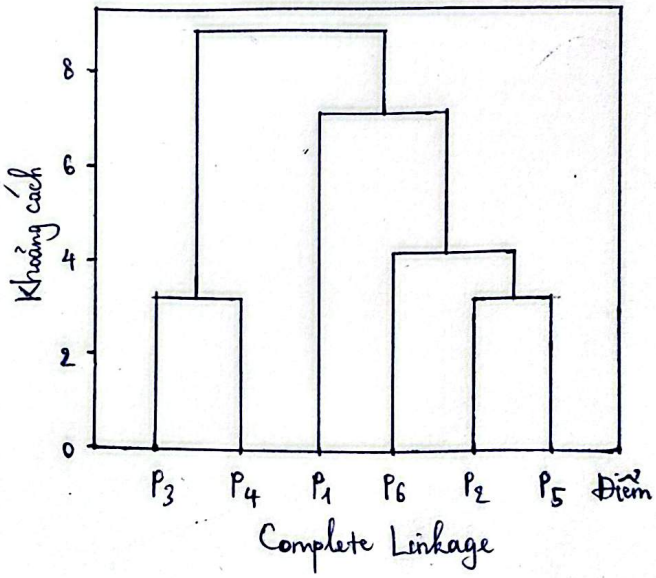
- Ma trận khoảng cách:

	P_1	P_2	P_3	P_4	P_5	P_6
P_1	0	3,61	7,28	6,08	6,08	7,07
P_2	3,61	0	5,10	5,66	3,16	3,61
P_3	7,28	5,10	0	3,16	7,21	3,61
P_4	6,08	5,66	3,16	0	8,60	6,08
P_5	6,08	3,16	7,21	8,60	0	4,12
P_6	7,07	3,61	3,61	6,08	4,12	0

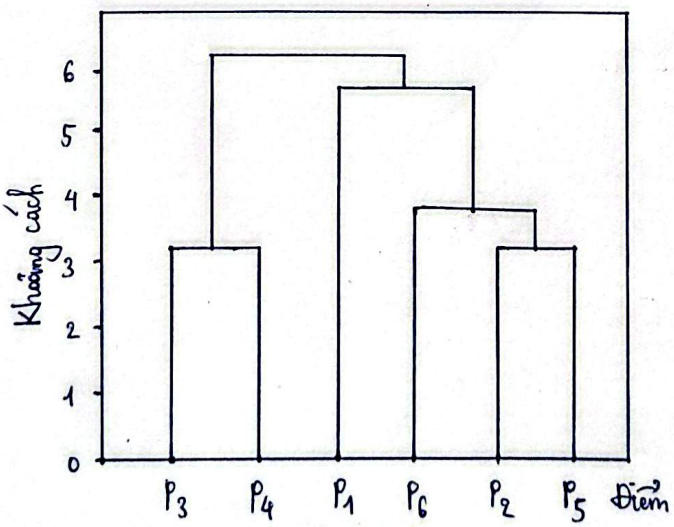
- Vẽ dendrogram bằng 3 phương pháp:



Single Linkage



Complete Linkage



Average Linkage

Được quét bằng CamScanner

d) A(4;4); B(9;3); C(5;9); D(3;7); E(8;8); F(6;2); G(10;6)

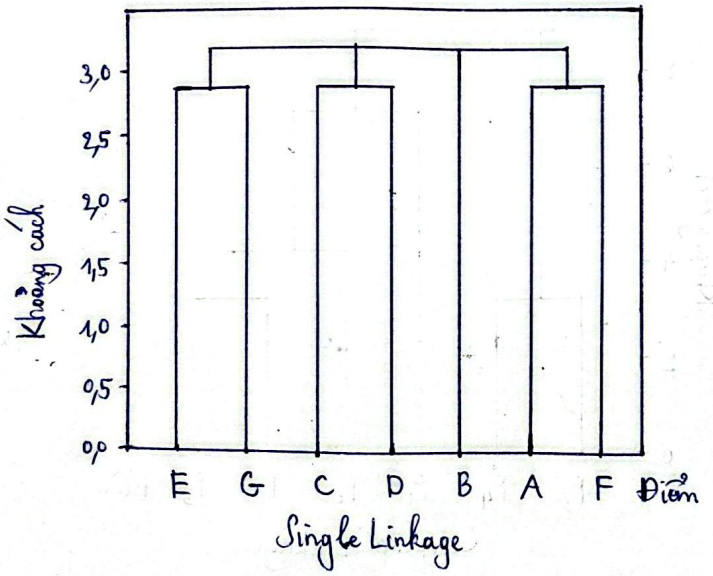
- Tính khoảng cách giữa các điểm, sử dụng công thức:

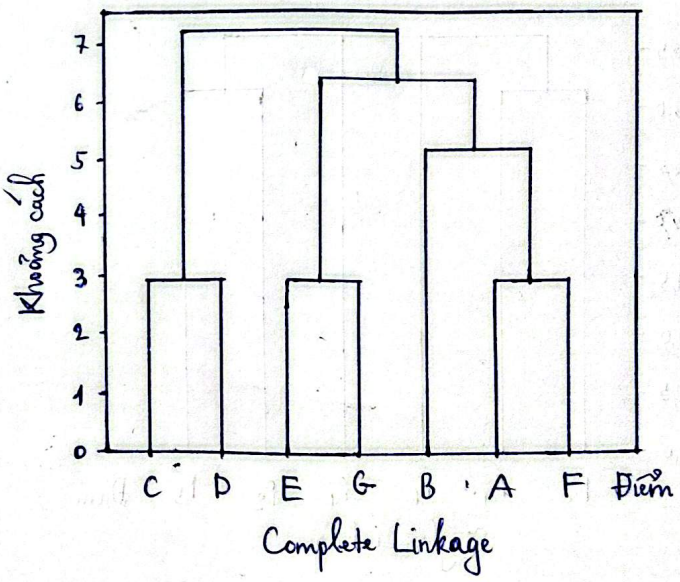
$$d(P_1, P_2) = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

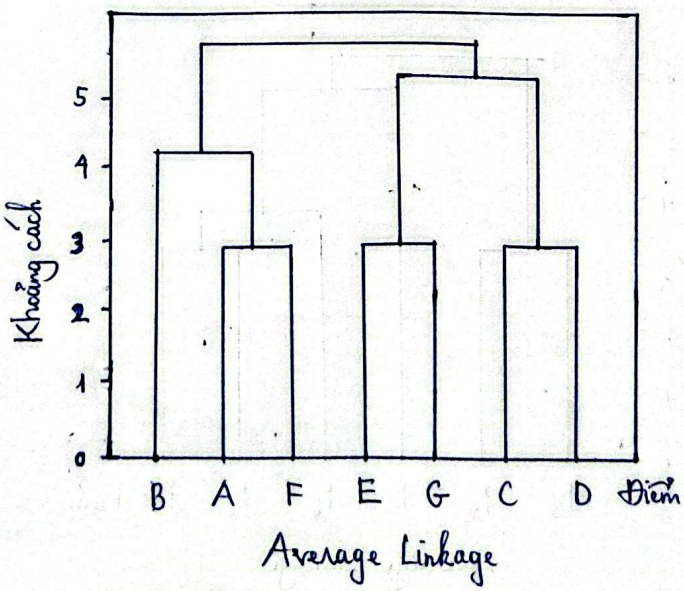
- Ma trận khoảng cách:

	A	B	C	D	E	F	G
A	0	5,10	5,10	3,16	5,66	2,83	6,32
B	5,10	0	7,21	7,21	5,10	3,16	3,16
C	5,10	7,21	0	2,83	3,16	7,07	5,83
D	3,16	7,21	2,83	0	5,10	5,83	7,07
E	5,66	5,10	3,16	5,10	0	6,32	2,83
F	2,83	3,16	7,07	5,83	6,32	0	5,66
G	6,32	3,16	5,83	7,07	2,83	5,66	0

- Vẽ dendrogram bằng 3 phương pháp:







Được quét bằng CamScanner